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DisSModel: A Python Framework for Spatially Explicit Dynamic Modeling

Sérgio Souza Costa [1], **Nerval de Jesus Santos Junior** [1], **Felipe Martins Sousa** [1], **and Denilson da Silva Bezerra** [1]

1 Universidade Federal do Maranhão (UFMA)

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- @HaoyuanHe0606
- @QRMM

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Summary

DisSModel (Discrete Spatial Modeling) is a modular Python framework for spatially explicit dynamic modeling, targeting Land Use and Land Cover Change (LUCC). It adapts the modeling paradigm of TerraME (Carneiro et al., 2013) to the Python ecosystem, enabling researchers to simulate socio-environmental systems — forest fires, epidemics, coastal dynamics — by coupling a time-stepped clock with the spatial data structures of GeoPandas (Jordahl et al., 2026).

Its **dual-substrate architecture** pairs a vector substrate (GeoDataFrame) for spatial expressiveness with a raster substrate (NumPy 2D arrays) for high-performance vectorized computation. DisSModel is available through the DisSModel GitHub organization and on PyPI.

Statement of Need

Python has become the lingua franca for geospatial data science, supported by libraries such as GeoPandas and PySAL — but these tools target static analysis. Dynamic spatial modeling, simulating how landscapes evolve over time, has historically required specialized platforms. In Brazil, TerraME (Carneiro et al., 2013) and Dinamica EGO (B. Soares-Filho et al., 2013; B. S. Soares-Filho et al., 2002) are the most widely adopted general-purpose frameworks, while narrower allocation models such as CLUE and CLUE-S (Veldkamp & Fresco, 1996; Verburg et al., 2002) are used elsewhere. Researchers working in Python thus lack a general-purpose option: the established frameworks are Lua- or GUI-based, and allocation models are usually standalone implementations without a shared interface.

TerraME is conceptually robust, but its reliance on Lua — far less common in data science than Python — is a barrier, and the framework has seen no new release since August 2020. General-purpose simulation libraries, meanwhile, lack native synchronization between a time-stepped clock and the state of a GeoDataFrame. DisSModel fills this gap with a lightweight scheduler coupled to vector and raster spatial state: an actively maintained Python implementation of the TerraME paradigm and, through its satellite package `dissluc`, of LuccME's CLUE-like and CLUE-S-like allocation components.

Reproducibility is a first-class concern: the `executor` module provides a standardized lifecycle — `validate → load → run → save` — capturing provenance metadata (input checksums, parameters, timing, output paths) in an `ExperimentRecord` generated automatically for every run, with no additional instrumentation by the modeler.

State of the Field

DisSModel occupies a niche between general-purpose agent-based modeling (ABM) libraries and specialized GIS simulation software:

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Aspect	TerraME	Dinamica EGO	DisSModel
Language	Lua	Visual/Internal	Python
Simulation Engine	Discrete Event	Cellular Automata	Time-stepped scheduler
Spatial Structure	CellularSpace (Fixed)	Cellular Grid	GeoDataFrame + NumPy (Dual)
GIS Integration	TerraLib	Native Raster	GeoPandas / Rasterio
Extensibility	Script-based	Block-based	Class Inheritance
Reproducibility	Manual	Manual	Automated (ExperimentRecord)

NetLogo (Wilensky, 1999) and Mesa (Kazil et al., 2020) are excellent for ABM; Mesa-Geo (Wang et al., 2022) adds GIS data to Mesa, whereas DisSModel centers on cellular-space state with a synchronized clock and a reproducible executor lifecycle. DisSModel builds on GeoPandas, following the discrete spatial modeling approach of Santos Junior (2025).

Software Design

DisSModel is organized into five modules with strict separation of concerns, extensible through class inheritance. **Core** manages the simulation clock: the `Environment` orchestrates time, and models auto-register at instantiation, run `setup` once, and receive each tick through `pre_execute` / `execute` / `post_execute` hooks. **Geo** provides the dual-substrate design: a vector substrate (`SpatialModel`, `CellularAutomaton`) on GeoDataFrame with libpysal neighborhoods (Rey et al., 2021), and a raster substrate (`RasterModel`, `RasterCellularAutomaton`) on NumPy arrays with vectorized operations (`shift2d`, `focal_sum`, `neighbor_contact`) replacing cell-by-cell loops. `SyncSpatialModel` and `SyncRasterModel` mirror TerraME's `synchronize()`, so coupled models sharing one grid read the same start-of-step state. **Executor** defines the `ModelExecutor` four-phase lifecycle (`validate`, `load`, `run`, `save`); subclasses self-register, and every run produces an `ExperimentRecord` with input checksum, parameters, timing, and output paths. **IO** provides a unified dataset abstraction across GeoDataFrame, GeoTIFF, and Xarray/Zarr, with transparent `s3://` resolution. **Visualization** integrates Matplotlib, Streamlit-compatible widgets, and `RasterMap`.

This extensibility has produced independent domain packages: `dissmodel-ca` (Costa, Santos Junior, & Zhao, 2026) (Cellular Automata), `dissmodel-sysdyn` (Costa, Santos Junior, & Vale, 2026) (System Dynamics), and `disslucc` (Costa, 2026b), which implements LuccME's continuous and discrete Demand, Potential, and Allocation components (Veldkamp & Fresco, 1996; Verburg et al., 2002) on the raster substrate and the same `ModelExecutor` contract, an explicit Python counterpart to TerraME/LuccME.

Validation and Performance

All benchmarks ran on an Intel Core i7-7700T @ 2.90GHz, 15 GB RAM (Linux Mint 22.2, Python 3.12.3, NumPy 2.5.3, GeoPandas 1.2.0); absolute timings vary by hardware.

Conway's Game of Life compares the two substrates with the same rule and initial state. The vector rule is a per-cell Python method querying libpysal Queen neighbors; the raster rule is one whole-grid focal sum; final states are compared cell by cell. At 100×100 cells the raster takes 0.34 ms/step against 3,783 ms/step for the vector (a gap that reflects the per-cell style of the vector API) and scales to 10 cells at 26.6 ms/step.

BR-MANGUE coastal dynamics. The coupled mangrove–flood model was established by Bezerra et al. (Bezerra et al., 2013) and extended in Bezerra et al. (2025), co-authored by D.S.B., F.M.S. and the submitting author, who are also responsible for the

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DisSModel reimplementation. `brmangue-dissmodel` (Bezerra et al., 2026) validates the raster implementation against a TerraME 2.0 adaptation of the published model (Bezerra & Costa, 2026) over the Maranhão Island dataset (50,496 cells); fidelity to the thesis is documented in the repository. Figures refer to `brmangue-dissmodel` v0.5.0. Match percentage suits categorical outputs (Pontius Jr. et al., 2011). In the baseline scenario (19 steps) land use and soil match exactly at every checkpoint and elevation matches on 97.4% of cells within 1 mm (MAE 0.00038 m; maximum error 0.10 m). In the flooding scenario (laboratory parameters) TerraME floods 2,469 non-sea cells by step 11 and the Python model 2,467; soil matches exactly, land use differs in 4 cells (0.008%), and at step 11 elevation matches on 94.4% of cells (MAE 0.008 m, maximum error 1.0 m). The residuals come from exact ties in elevation: the flux rule compares accumulated elevations with <=, so a ~1e-16 difference in summation order (sequential in TerraME, per direction in NumPy) can flip a comparison and, near the flooding threshold, a cell's class. They are floating-point effects, not a difference in the rules. Both scenarios are reproducible with the validation executor against golden files shipped with the package, which the adaptation's headless driver regenerates (Bezerra & Costa, 2026).

Cross-substrate equivalence (60×60 synthetic grid, 10 steps) gives 100% match for land use, soil and elevation within a 5 cm tolerance (MAE 0.0011 m, max 0.024 m; floating-point rounding), with raster at 2.5 ms/step against 69.1 ms/step for vector (28.1×; the vector port follows TerraME's per-cell loops).

disslucc as a second case. `disslucc` (Costa, 2026b), built on DisSModel, reimplements the LuccME demand, potential and allocation components (continuous CLUE-like (Veldkamp & Fresco, 1996) and discrete CLUE-S-like). It tests the `ModelExecutor` contract and the raster substrate against a second TerraME application, not as a result in itself. Its checks live in a separate, versioned benchmark (Costa, 2026c) that compares every simulated year, iteration counts included, with per-year reference outputs (Costa, 2026d) generated in a containerized TerraME (Costa, 2026e) and checked by SHA-256. Scenarios follow the LuccME test labs (`labNN`); the `labNN_mdX` variants exercise the convergence loop. MAE suits fractional outputs (Pontius Jr. et al., 2011; Willmott & Matsuura, 2005).

`lab01_md1643` (continuous, 6,574 cells, 2008–2014) differs from the reference by MAE = 0.0036 in the final year ([0,1] scale) with `disslucc`'s defaults. The difference is one deliberate deviation: LuccME's `correctCellChange` never executes (its guard reads a mismatched field name), while `disslucc` runs it by default. With `cell_correction=False` (the benchmark's `lab01_md1643` scenario) the match holds in every year, with identical iteration counts (MAE < 1e-7). `lab15_md10` (discrete, 5,914 cells) reproduces the reference cell for cell, with zero quantity and allocation disagreement (Pontius Jr. & Millones, 2011) and identical iteration counts. Overall, ten of 21 LuccME test labs and both variants match with identical iteration counts and differences up to 3e-6 (single precision); a validated lab runs in 0.11 to 0.55 s on the hardware above (median of five; `make timing`). The other eleven labs need potential and allocation components not yet implemented; student fellows are developing them, and each lab joins the benchmark as its components land.

Research Impact Statement

DisSModel's scientific lineage is rooted in the TerraME/LuccME research program at INPE, where TerraME was developed by Tiago Carneiro (Carneiro et al., 2013) and LuccME by Ana Paula Dutra Aguiar (Aguiar et al., 2019), both former doctoral students of Prof. Gilberto Câmara. The submitting author conducted doctoral research in the same program, under Câmara and Aguiar, and has co-authored its modeling work since 2009 (Costa et al., 2009; Moreira et al., 2009); `disslucc` reimplements in Python the continuous and discrete allocation components of that lineage. DisSModel was presented at a seminar of INPE's Applied Computing graduate program (CAP) on 7 May 2026 (Costa, 2026f).

The framework is in use across two UFMA research groups. Within LambdaGeo, `brmangue-dissmodel` builds on the reference implementation by co-author F.M.S. Co-author D.S.B.

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(former INPE), co-leader of UFMA's Geotechnologies and Environmental Sciences Laboratory (GEOTAM), whose doctoral work founded BR-MANGUE (Bezerra et al., 2013), also adopts the DisSModel reimplementation in his coastal dynamics projects (UFMA PVCBS4959-2025 and PVCBS4960-2025), a collaboration predating DisSModel itself (Bezerra et al., 2025).

Since September 2026, four undergraduate research fellows (UFMA, CNPq) are being trained on the project; the stabilized `ModelExecutor` contract lets each own an independent repository — `dissluccc`, `brmangue-dissmodel`, or `disscube` (Costa, 2026a) (a data-cube layer, a Python alternative to TerraME's `fillCellularSpace`) — without core changes.

A roadmap toward DisSModel 1.0 (May 2027) includes an open textbook, *Geospatial Modeling with Python* (https://lambdageo.github.io/geospatial-modeling-python/), already in progress. On the data side, `disscube` already reads Brazil Data Cube collections through their STAC catalog (Ferreira et al., 2020), so DisSModel models can run on the same Earth observation data used by SITS (Simoes et al., 2021).

Author Contributions

CRediT roles: **S.S.C.** — Conceptualization, Software (Core, Geo, Executor, IO, Visualization), Methodology, Validation, Writing, Supervision, Project administration. **N.J.S.J.** — Conceptualization, Software (initial design), Validation, Writing (undergraduate thesis (Santos Junior, 2025)). **D.S.B.** — Conceptualization (domain science), Validation, Resources (Bezerra et al., 2013, 2025). **F.M.S.** — Conceptualization, Software (BR-MANGUE reference implementation), Data curation, Validation (Bezerra et al., 2025). All authors reviewed and approved the final manuscript.

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AI Usage Disclosure

The initial prototype (May–June 2025), the second author's undergraduate thesis (Santos Junior, 2025), did not involve generative AI. From February 2026, Claude and Gemini CLI supported documentation, tests and refactoring of existing code. Since mid-2026, Claude Code has been used across the whole ecosystem (framework, satellite packages and benchmarks) for coding, bug fixing, refactoring, code translation, tests, validation scripts, documentation, and this paper's text. This made the ecosystem envisioned by the submitting author feasible. The reference outputs used in validation are produced by TerraME itself, running the model files in a container; AI-assisted changes to those files were checked to leave the outputs unchanged. The scientific design — TerraME compatibility contract, executor pattern, dual-substrate architecture, validation methodology — is the authors', and traces to the submitting author's doctoral research at INPE. The authors reviewed, tested and ran all AI-generated code and text.

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